Maria A. Gutierrez

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Education

Oct 2021- Sep 2025 **PhD in Applied Mathematics, Gates Cambridge Scholar**, *University of Cambridge (UK)*
Thesis "Population dynamics of infectious disease and pathogen evolution: modelling vaccine escape": link
Supervisor (US equivalent: advisor): Julia Gog OBE; Advisor (secondary supervisor): Michael Cates FRS

2020-2021 **Mathematics MMath (Part III, Distinction)**, *University of Cambridge (UK)*

2017-2020 **Mathematics BA (First Class Honours)**, *University of Cambridge (UK)*

Research experience

Oct 2025- **Postdoctoral Research Associate**, *Odum School of Ecology, University of Georgia (USA)*, Rohani Lab, ecology and evolution of influenza viruses

2021 **Quantitative finance summer internship**, *Susquehanna Intl. Group, Dublin (Ireland)*

2020 **Cosmology summer project**, *University of Cambridge (UK)*, with Enrico Pajer

2019 **Biophysics summer project**, *U. of Cambridge*, with Pierre Haas and Raymond Goldstein FRS

Publications (Google Scholar)

2026 **Identifying the viral and epidemiological factors behind the apparent global extinction of influenza B/Yamagata**, under review, Gutierrez MA and Rohani P: preprint link

2026 **Strain coexistence and competition for pathogens with asymmetric cross-immunity and waning immunity**, under review, Gutierrez MA, Page CK, Tompkins SM, Rohani P: preprint link

2026 **Epidemic dynamics shape variant appearance and stochastic establishment: implications for vaccination**, under review, Gutierrez MA and Gog JR: preprint link

2026 **Final-size solutions for SIRI models with vaccination**, *Bulletin of Mathematical Biology*, 88,64, Gutierrez MA and Gog JR: link

2026 **Infectious disease outbreak controllability: biological, social and public health factors**, *Proceedings of the Royal Society B: Biological Sciences*, 293 (2063): 20252848, Thompson RN, Bansal S, Clapman H, Dyson L, Gutierrez MA, *et al.*: link

2023 **The importance of vaccinated individuals to population-level evolution of pathogens**, *Journal of Theoretical Biology*, 567, 111493, Gutierrez MA and Gog JR: link

2022 **Stabilization of microbial communities by responsive phenotypic switching**, *Physical Review Research*, 4, 033224, Haas PA, Gutierrez MA, Oliveira NM, Goldstein RE: link

Awards and grants

2026 **PhD thesis prize (£500)**, *Cambridge Centre for Physical Biology*

2021-2025 **Gates Cambridge scholarship**, *Bill & Melinda Gates Foundation*, merit, service and leadership full scholarship (stipend + international fees) for outstanding international PhD candidates

2022-2025 **Academic travel grants**, *European Society of Mathematical and Theoretical Biology; King's College (U. of Cambridge); Gates Cambridge Trust*, to Germany, Finland, Sweden, Italy, Portugal

2021 **Claredon scholarship (declined)**, *University of Oxford*, full scholarship for PhD position

2021 **Roth scholarship (declined)**, *Imperial College London*, full scholarship for PhD position

2018-2021 **Examination prizes**, *Christ's College (U. of Cambridge)*, for outstanding academic performance

2017 **Excellence scholarship (2500€)**, *Region of Madrid, Department of Education*

2017 **Entrance award (500€)**, *Universidad Complutense de Madrid*

2017 **Honours scholarship**, *Sagrado Corazon Madrid School*, **over 1000€** in university fees

Research presentations

Upcoming **International Centre for Applied Mathematical Modelling and Data Analytics**, webinar

Upcoming **JUNIPER Partnership**, *United Kingdom*, webinar

2026 **CIDER (Center for Influenza Disease & Emergence Research) flu day**, *University of Georgia*, talk: slides

2026 **CEIRR (Centers of Excellence for Influenza Research and Response) Network annual meeting**, *Emory University*, poster and flash talk: pdf

2026 **Center for the Ecology of Infectious Diseases annual retreat**, *University of Georgia*, talk

2026 **Cambridge CPB (Centre for Physical Biology) annual meeting**, *University of Cambridge*, prize talk (invited): slides

2025 **EEID25 (Ecology and Evolution of Infectious Diseases) conference**, *University of Notre Dame*, poster: pdf

2025 **Bifurcation and Chaos in Population Biology workshop**, *Brin Mathematics Research Center (University of Maryland)*, invited speaker: slides

2025 **MPDEE25 (Conference on Models in Population Dynamics, Ecology, and Evolution)**, selected talk: slides

2024 **VGEB24 (Virus, Genomics, Evolution & Bioinformatics) conference**, poster: pdf, video

2024 **MIDAS (Models of Infectious Disease Agent Study) network**, webinar: slides, video

2024 **Isaac Newton Institute for Mathematical Sciences: Modelling and inference for pandemic preparedness (workshop)**, *University of Cambridge*, selected talk: slides, video

2024 **Centre for Ecology and Evolution (Imperial/UCL) Spring Symposium: The evolution of viral epidemics and pandemics: past, present and future**, *London*, invited speaker: slides

2024 **Disease Dynamics and Pathogen Genomics seminar**, *Department of Veterinary Medicine, University of Cambridge*, slides

2024 **SMB MathEpiOnco 2024 (online mini conference)**, selected talk: slides

2024 **DSABNS 2024 (15th Conference on Dynamical Systems Applied to Biology and Natural Sciences)**, *Lisbon (Portugal)*, invited lecture: slides

2023 **EPIDEMICS9 (9th International Conference on Infectious Disease Dynamics)**, *Bologna (Italy)*, selected talk: slides

2023 **IDDConf2023 (Infectious Diseases Dynamics Conference)**, *Lake District (UK)*, talk: slides

2023 **UNAM, Faculty of Mathematics**, *Mexico City (Mexico)*, seminar in Spanish: slides

2023 **NORDITA program: Unifying the epidemiological and evolutionary dynamics of pathogens**, *Stockholm (Sweden)*, research seminar: slides

2023 **Cambridge Infectious Diseases Annual Symposium**, *Cambridge (UK)*, selected talk: slides

2023 **JUNIPER Consortium**, *United Kingdom*, online seminar: slides

2022 **ECMTB 2022 (SMB - ESMTB joint meeting: 12th European Conference on Mathematical and Theoretical Biology)**, *Heidelberg (Germany)*, selected poster: pdf

2022 **Cambridge Infectious Diseases Annual Symposium**, *Cambridge (UK)*, poster: pdf

Teaching

2021-2024 **Invited lecturer**, *Introduction to Mathematical Biology*, for junior high school students
Christ's - Trinity Women in Maths Residential 2021, 2024; King's - Christ's Joint Easter Residential 2023

2021-2023 **Undergraduate supervisor**, *Differential Equations, Methods, Mathematical Biology*

2022 **Undegraduate mathematics admissions interviewer**, *King's College Cambridge*

2021-2022 **Drop-in sessions (office hours for master courses)**, *Stochastic Processes in Biology*

Public talks and outreach activities

2025-2026 **Series of 30 Youtube videos**, sharing my experience and advice on academia, link

2024 **Mathematicians should write (written article)**, *Eureka (issue 67)*, link

2024 **LEGO epidemics**, *Cambridge Science Festival*, hands-on activity for children

2023 **Maths and viruses**, *50 Years of Women at King's Festival*, invited speaker: slides

2023 **Contagious maths (for young children)**, *'Meet the researchers'*, video

2022 **Maths and the evolution of viruses**, *Christ's College Maths Society*, invited speaker: slides

2022 **A mathematician's adventures in pandemic modelling**, *Pint of Science (international popular science festival)*, invited speaker: slides

2021 **Statistical field theory in biological systems**, *Part III Seminar Series*, talks by masters students

2020 **Mathematical epidemiology**, *Cambridge University Biological Society, The Archimedeans (Cambridge University Mathematical Society)*, student talks for and by undergraduates

2019 **Una matemática biológica**, *Sagrado Corazon Madrid School*, for senior high school students

Service

2023- **Peer-reviewing**, *PNAS, Jour. of Theoretical Biology, Bull. of Mathematical Biology, Ecosphere*

2019-2022 **Mentoring of female students**, high-schoolers from Spain and Cambridge undegraduates

Committees

2023-2024 **King's College Boat Club**, alumni officer
Organised a boat naming ceremony and the alumni boat club dinner, held at King's College Cambridge

2022-2023 **Equality, Diversity & Inclusion (Faculty of Mathematics)**, graduate representative
Arranged for free menstrual products, LGBT/Trans flags during pride month/trans awareness week

2021-2022 **Emmy Noether Society (for women in mathematics)**, graduate representative
Gave two study skills workshops, organised socials, and talks by female academics

2020-2021 **Effective Altruism Cambridge**, consultant
Led 3 online careers workshops (over 200 sign-ups each) on "High Impact Careers for Mathematicians"